Trout egg survival: overdispersion and quasilikelihood

AUTHORS
Patricia Menéndez
Heejung Shim

Trout egg survival data

Boxes of trout eggs were buried at five different stream locations and retrieved at four different times. For each combination of location and period, the number of surviving eggs was recorded.

Our question of interest is:

> **Does the probability of survival of trout eggs depend on location and/or period?**

For observation i, let

$$Y_i = \text{number of surviving eggs},$$

and let m_i denote the total number of eggs in that box. Under an ordinary binomial model,

$$Y_i \sim \text{Binomial}(m_i, p_i),$$

so that

$$E(Y_i) = m_i p_i = \mu_i,$$

and

$$\text{Var}(Y_i) = m_i p_i (1 - p_i) = \frac{\mu_i (m_i - \mu_i)}{m_i}.$$

We will start with this binomial model and then check whether its variance assumption is reasonable for these data.

```
library(faraway)
library(tidyverse)
data(troutegg)
?troutegg
str(troutegg)
```

```
'data.frame':   20 obs. of  4 variables:
 $ survive : int  89 106 119 104 49 94 91 100 80 11 ...
 $ total   : int  94 108 123 104 93 98 106 130 97 113 ...
 $ location: Factor w/ 5 levels "1","2","3","4",..: 1 2 3 4 5 1 2 3 4 5 ...
 $ period  : Factor w/ 4 levels "4","7","8","11": 1 1 1 1 1 2 2 2 2 2 ...
```

Exploring the data

We first look at the observed numbers of surviving eggs and the corresponding totals for each location and period.

```
ftable(xtabs(cbind(survive,total) ~ location + period, troutegg))
```

```
                survive total
location period
1        4            89    94
         7            94    98
         8            77    86
         11          141   155
2        4           106   108
         7            91   106
         8            87    96
         11          104   122
3        4           119   123
         7           100   130
         8            88   119
         11           91   125
4        4           104   104
         7            80    97
         8            67    99
         11          111   132
5        4            49    93
         7            11   113
         8            18    88
         11            0   138
```

```
troutegg |>
  select(location, period, survive, total) |>
  arrange(location, period)
```

```
   location period survive total
1         1      4      89    94
6         1      7      94    98
11        1      8      77    86
16        1     11     141   155
2         2      4     106   108
7         2      7      91   106
12        2      8      87    96
17        2     11     104   122
3         3      4     119   123
8         3      7     100   130
13        3      8      88   119
18        3     11      91   125
4         4      4     104   104
9         4      7      80    97
14        4      8      67    99
19        4     11     111   132
5         5      4      49    93
10        5      7      11   113
15        5      8      18    88
20        5     11       0   138
```

It is also useful to look at the observed survival proportions. This gives us a direct view of how survival changes across locations and periods before fitting a model.

```
troutegg <- troutegg |>
  mutate(
    prop_survive = survive / total,
    fail = total - survive
  )
```

```
troutegg |>
  select(location, period, survive, fail, total, prop_survive)
```

```
   location period survive fail total prop_survive
1         1      4      89    5    94   0.94680851
2         2      4     106    2   108   0.98148148
3         3      4     119    4   123   0.96747967
4         4      4     104    0   104   1.00000000
5         5      4      49   44    93   0.52688172
6         1      7      94    4    98   0.95918367
7         2      7      91   15   106   0.85849057
8         3      7     100   30   130   0.76923077
9         4      7      80   17    97   0.82474227
10        5      7      11  102   113   0.09734513
11        1      8      77    9    86   0.89534884
12        2      8      87    9    96   0.90625000
13        3      8      88   31   119   0.73949580
14        4      8      67   32    99   0.67676768
15        5      8      18   70    88   0.20454545
16        1     11     141   14   155   0.90967742
17        2     11     104   18   122   0.85245902
18        3     11      91   34   125   0.72800000
19        4     11     111   21   132   0.84090909
20        5     11       0  138   138   0.00000000
```

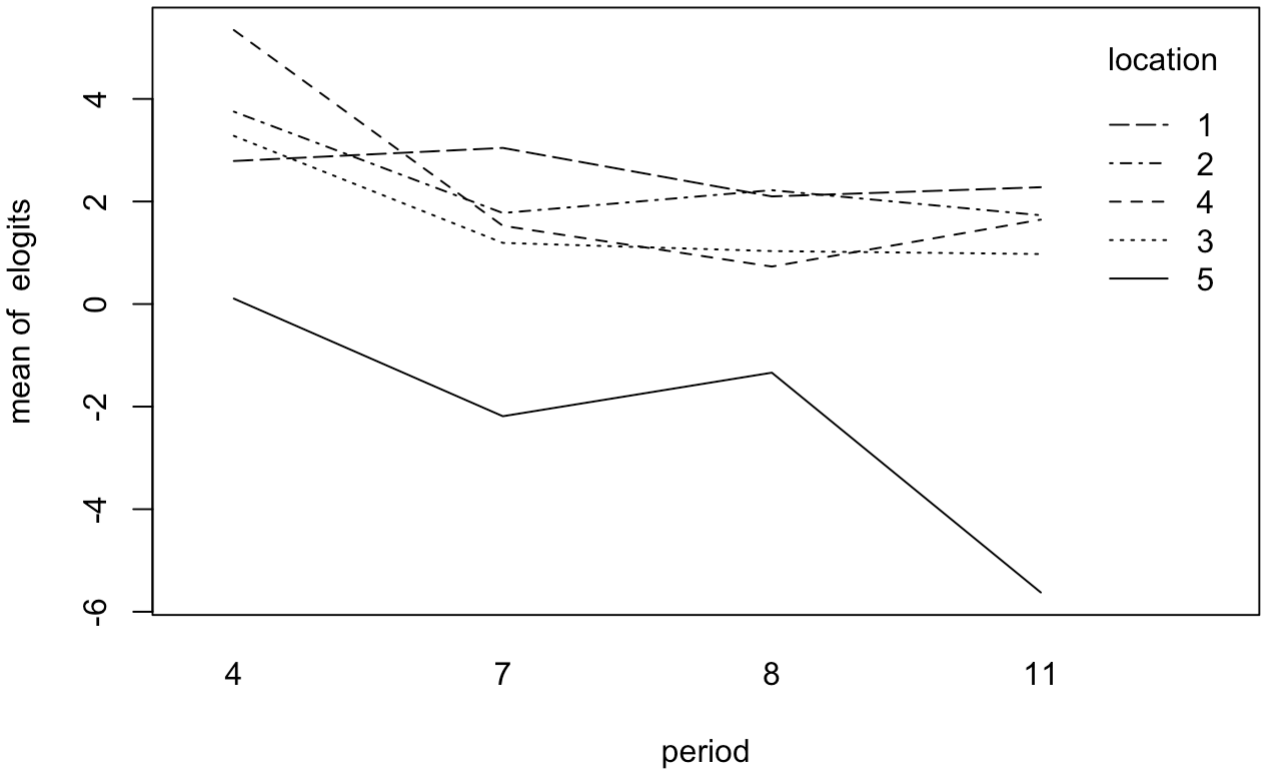
Before fitting an additive model, it is useful to check whether there is evidence of an interaction between `location` and `period`.

For a binomial response, the model will use the logit link,

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \eta_i.$$

The following code constructs empirical logits. The addition of 0.5 is a small correction that avoids problems if the observed number of successes or failures is zero.

```
elogits <- log((troutegg$survive+0.5)/(troutegg$total-troutegg$survive+0.5))
with(troutegg, interaction.plot(period, location, elogits))
```




```
# No obvious interaction is visible.
```

If the profiles are reasonably similar across locations, an additive model with main effects for `location` and `period` is a sensible starting point:

$$\text{logit}(p_i) = \mathbf{x}_i^T \boldsymbol{\beta}.$$

Fitting an ordinary binomial regression

We first fit the standard binomial GLM with a logit link:

```
bmod <- glm(cbind(survive,total-survive) ~ location+period,
            family=binomial, troutegg)
summary(bmod)
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ location + period,
    family = binomial, data = troutegg)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   4.6358     0.2813  16.479  < 2e-16 ***
location2    -0.4168     0.2461  -1.694   0.0903 .
location3    -1.2421     0.2194  -5.660 1.51e-08 ***
location4    -0.9509     0.2288  -4.157 3.23e-05 ***
location5    -4.6138     0.2502 -18.439  < 2e-16 ***
period7      -2.1702     0.2384  -9.103  < 2e-16 ***
```

```
period8      -2.3256     0.2429  -9.573  < 2e-16 ***
period11     -2.4500     0.2341 -10.466  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1021.469  on 19  degrees of freedom
Residual deviance:   64.495  on 12  degrees of freedom
AIC: 157.03

Number of Fisher Scoring iterations: 5
```

A reminder about the deviance reported by `glm()`

The `glm()` output reports both a **null deviance** and a **residual deviance**. They use the same deviance definition introduced in class, but correspond to two different fitted models.

The **null deviance** is the deviance of the intercept-only model,

$$g(\mu_i) = \beta_0,$$

so

$$D_{\text{null}} = 2\,\{\ell(\text{saturated}) - \ell(\text{intercept-only})\}.$$

The **residual deviance** is the deviance of the model that we have actually fitted,

$$D_{\text{res}} = 2\,\{\ell(\text{saturated}) - \ell(\text{fitted})\}.$$

Therefore,

$$D_{\text{null}} - D_{\text{res}}$$

measures how much the predictors improve the fit relative to the intercept-only model.

For checking the adequacy of the fitted model, however, it is the **residual deviance** that we compare with its residual degrees of freedom.

For an ordinary binomial model the dispersion parameter is fixed at

$$\phi = 1.$$

Therefore the model assumes

$$\text{Var}(Y_i) = \frac{\mu_i(m_i - \mu_i)}{m_i}.$$

Is the binomial model adequate?

The residual deviance compares the fitted model with the saturated model. Under the usual large-sample approximation, if the binomial model is adequate,

$$D \approx \chi^2_{n-p},$$

where n is the number of observations and p is the number of fitted regression parameters.

Here there are $n = 20$ observations and $p = 8$ regression parameters, so

$$n - p = 12.$$

```
deviance(bmod)
```

```
[1] 64.49512
```

```
df.residual(bmod)
```

```
[1] 12
```

```
pchisq(
  deviance(bmod),
  df = df.residual(bmod),
  lower.tail = FALSE
)
```

```
[1] 3.379416e-09
```

This computes

$$P\left(\chi^2_{12} \geq D_{\text{obs}}\right),$$

where D_{obs} is the residual deviance from the fitted binomial model.

In other words, we are asking:

> If the fitted binomial GLM were adequate, how likely would we be to observe a residual deviance at least this large?

The hypotheses can be viewed as

$$H_0: \quad \text{the fitted binomial GLM is adequate},$$

against lack of fit of the model.

A very small p-value therefore provides evidence that the ordinary binomial model does not adequately describe the data.

Importantly, this test identifies **lack of fit**, but it does not tell us the reason for that lack of fit. Possible causes include:

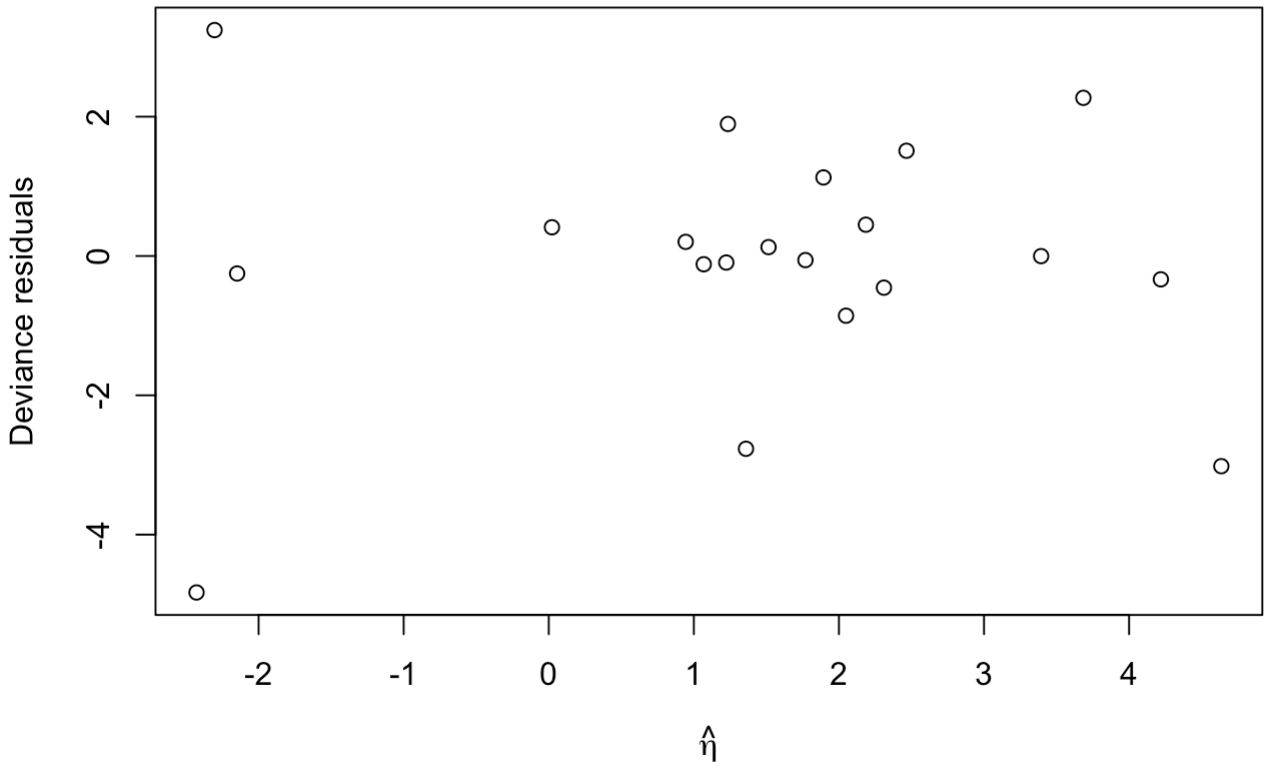
- overdispersion;
- an incorrect mean structure;
- a missing interaction.

For this reason, we should inspect the residuals before deciding that overdispersion is the main problem.

Checking residuals

Before concluding that the problem is overdispersion, we should also check whether there is an obvious systematic pattern in the residuals.

```
plot(residuals(bmod) ~ predict(bmod,type="link"),
     xlab=expression(hat(eta)), ylab="Deviance residuals")
```



The absence of a clear trend suggests that the mean structure may be reasonable. The issue may instead be that the assumed binomial variance is too small.

This is an important distinction:

> The model for the **mean** may be appropriate even when the model for the **variance** is not.

Estimating the dispersion parameter

For a quasilikelihood model, instead of fixing $\phi = 1$, we allow

$$\mathrm{Var}(Y_i) = \phi v(\mu_i).$$

For a quasi-binomial model,

$$v(\mu_i) = \frac{\mu_i(m_i - \mu_i)}{m_i},$$

so

$$\boxed{\mathrm{Var}(Y_i) = \phi \frac{\mu_i(m_i - \mu_i)}{m_i}}.$$

If $\phi > 1$, the data are overdispersed relative to the ordinary binomial model.

A commonly used estimate of ϕ is based on Pearson's chi-squared statistic,

$$X^2 = \sum_{i=1}^{n} \frac{(y_i - \widehat{\mu}_i)^2}{v(\widehat{\mu}_i)},$$

with

$$\boxed{\widehat{\phi} = \frac{X^2}{n-p}}.$$

For an ordinary binomial model we would expect $\widehat{\phi}$ to be reasonably close to 1. A value substantially larger than 1 indicates overdispersion. There is **no universal cutoff** for deciding how large $\widehat{\phi}$ must be before overdispersion is important. Values close to 1 are consistent with the variance assumed by the ordinary binomial model, whereas values appreciably larger than 1 suggest overdispersion. The size of $\widehat{\phi}$ should be interpreted together with the residual diagnostics and the adequacy of the mean model.

```
n <- 20
p <- 8
(phihat <- sum(residuals(bmod,type="pearson")^2)/(n-p))
```

```
[1] 5.330322
```

The practical consequence is that overdispersion affects the **uncertainty** in the estimated regression coefficients. Under the quasi-binomial model,

$$\mathrm{SE}_{\mathrm{quasi}}(\widehat{\beta}_j) = \sqrt{\widehat{\phi}}\ \mathrm{SE}_{\phi=1}(\widehat{\beta}_j).$$

Thus, for example, if $\widehat{\phi} = 4$, the estimated variances of the coefficients are four times as large and their standard errors are twice as large.

Why quasilikelihood?

With maximum likelihood we specify a complete probability distribution for Y_i. Quasilikelihood allows us to work with less information. We specify the mean, variance and link function:

$$E(Y_i) = \mu_i,$$
$$\mathrm{Var}(Y_i) = \phi v(\mu_i),$$

and

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}.$$

We do **not** need to specify the full distribution of Y_i.

The corresponding quasi-score is

$$u_i = \frac{Y_i - \mu_i}{\phi v(\mu_i)}.$$

It has the important score-like properties

$$E(u_i) = 0,$$

and

$$-E\left(\frac{\partial u_i}{\partial \mu_i}\right) = E(u_i^2).$$

These are properties shared by an ordinary likelihood score. This motivates defining a quasi-log-likelihood contribution Q_i such that

$$\frac{\partial Q_i}{\partial \mu_i} = u_i.$$

Thus,

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi v(t)}\, dt.$$

The quasi-log-likelihood is then

$$Q = \sum_{i=1}^{n} Q_i.$$

The important point is that Q is not necessarily the logarithm of a genuine probability density. It is constructed to play the role of a log-likelihood for estimation and inference.

Fitting the quasi-binomial model

We now fit the same mean model using a quasi-binomial variance:

```
qbmod <- glm(cbind(survive,total-survive) ~ location+period,
             family=quasibinomial, troutegg)
summary(qbmod)
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ location + period,
    family = quasibinomial, data = troutegg)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.6358     0.6495   7.138 1.18e-05 ***
location2    -0.4168     0.5682  -0.734 0.477315
location3    -1.2421     0.5066  -2.452 0.030501 *
location4    -0.9509     0.5281  -1.800 0.096970 .
location5    -4.6138     0.5777  -7.987 3.82e-06 ***
period7      -2.1702     0.5504  -3.943 0.001953 **
period8      -2.3256     0.5609  -4.146 0.001356 **
period11     -2.4500     0.5405  -4.533 0.000686 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 5.330358)

    Null deviance: 1021.469  on 19  degrees of freedom
Residual deviance:   64.495  on 12  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

Why are the estimated coefficients the same?

For the quasi-binomial model, the quasi-log-likelihood has the same dependence on $\boldsymbol{\beta}$ as the ordinary binomial log-likelihood, apart from the common factor $1/\phi$.

Equivalently, the quasi-score equations for $\boldsymbol{\beta}$ contain the same common factor $1/\phi$. Since $\phi > 0$, multiplying an estimating equation by $1/\phi$ does not change the value of $\boldsymbol{\beta}$ that solves it.

Therefore,

$$\boxed{\widehat{\boldsymbol{\beta}}_{\text{quasi-binomial}} = \widehat{\boldsymbol{\beta}}_{\text{binomial}}}.$$

The fitted probabilities $\widehat{p}_i$ and fitted means $\widehat{\mu}_i$ are therefore also the same.

Where does ϕ matter?

Although ϕ does not change $\widehat{\boldsymbol{\beta}}$, it **does** change the uncertainty associated with those estimates.

The estimated covariance matrix is inflated by $\widehat{\phi}$:

$$\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}})_{\text{quasi}} = \widehat{\phi}\,\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}})_{\text{binomial}}.$$

Hence the standard errors are multiplied by

$$\boxed{\sqrt{\widehat{\phi}}}.$$

For example, the following calculation rescales one of the ordinary binomial standard errors by $\sqrt{\widehat{\phi}}$:

```
0.2461*sqrt(phihat)
```

```
[1] 0.5681831
```

So, when $\widehat{\phi} > 1$:

- the coefficient estimates stay the same;
- the standard errors become larger;
- confidence intervals become wider;
- test statistics generally become smaller in magnitude;
- fewer terms may appear statistically significant.

IWLS and quasilikelihood

The IWLS algorithm does not require a full likelihood. It uses the link function and the variance function.

For the quasi-binomial model,

$$\mathrm{Var}(Y_i) = \phi v(\mu_i).$$

The IWLS weights contain the factor $1/\phi$. Since this is a common factor for all observations, it cancels from the update for $\boldsymbol{\beta}$.

Thus, the same IWLS algorithm can be used to estimate $\boldsymbol{\beta}$. In the quasilikelihood setting, IWLS solves the **quasi-score equations** rather than ordinary likelihood score equations.

This explains why we do not need to replace a likelihood inside IWLS: IWLS itself only needs the mean, variance and link structure.

Remember

$$\text{same } \widehat{\boldsymbol{\beta}} \qquad \text{but} \qquad \text{larger uncertainty when } \widehat{\phi} > 1.$$

Quasideviance

Just as ordinary deviance compares a fitted model with the saturated model, we can define a quasideviance.

Recall that

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi v(t)}\, dt.$$

For the saturated model, $\mu_i = y_i$, and therefore

$$Q_i^s = \int_{y_i}^{y_i} \frac{y_i - t}{\phi v(t)}\, dt = 0.$$

So the **scaled quasideviance** is

$$\boxed{\frac{D_Q}{\phi} = -2 \sum_{i=1}^{n} Q_i}.$$

Removing the factor $1/\phi$ gives the **quasideviance**

$$\boxed{D_Q = -2 \sum_{i=1}^{n} \int_{y_i}^{\mu_i} \frac{y_i - t}{v(t)}\, dt}.$$

Thus,

$$\text{scaled quasideviance} = \frac{D_Q}{\phi}.$$

The quasideviance measures lack of fit, while the scaled quasideviance adjusts that lack of fit for the overall dispersion.

For quasi-binomial and quasi-Poisson models, the quasideviance D_Q has the same expression as the ordinary binomial or Poisson deviance. However, because ϕ is no longer fixed at 1, inference based on changes in deviance must account for the estimated dispersion.

Testing whether location is needed

We now compare two nested quasi-binomial models.

The **larger model** contains both location and period:

$$\mathcal{M}_B: \quad \text{logit}(p_i) = \text{location} + \text{period},$$

while the **smaller model** contains period only:

$$\mathcal{M}_A: \quad \text{logit}(p_i) = \text{period}.$$

The hypotheses are

$$H_0 : \text{the location terms are not needed},$$

against

$$H_1 : \text{at least one location term is needed}.$$

```
qbmod2 <- glm(cbind(survive,total-survive) ~ period,
              family=quasibinomial, troutegg)
summary(qbmod2)
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ period, family = quasibinomial,
    data = troutegg)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.139      1.070   1.999   0.0629 .
period7       -1.333      1.277  -1.044   0.3118
period8       -1.336      1.298  -1.029   0.3186
period11      -1.453      1.233  -1.178   0.2561
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 56.33299)

    Null deviance: 1021.47  on 19  degrees of freedom
Residual deviance:  913.56  on 16  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

Why do we use an F test?

For ordinary binomial and Poisson GLMs, $\phi = 1$, and differences in deviance are commonly compared using a χ^2 reference distribution.

For a quasi model, ϕ is unknown and must be estimated. If model A is nested within model B, and model B has s additional parameters, the quasi-likelihood test statistic is

$$\boxed{F_Q = \frac{(D_Q^A - D_Q^B)/s}{\widehat{\phi}} \approx F_{s,n-p}}.$$

Here:

- D_Q^A is the quasideviance of the smaller model;
- D_Q^B is the quasideviance of the larger model;
- s is the difference in the number of parameters;

- p is the number of parameters in the larger model;
- $\widehat{\phi}$ is estimated from the larger model.

For this example, adding `location` contributes four additional parameters, so

$$s = 4,$$

and the larger model has $p = 8$ parameters. Therefore the reference distribution is

$$F_{4,12}.$$

The calculation performed by the model comparison can be checked manually:

```
# Check calculation of the F statistic for testing location
(913.56 - 64.5)/4/phihat
```

```
[1] 39.82218
```

The numerator

$$D_Q^A - D_Q^B$$

measures the improvement in fit obtained by adding `location`. We divide by s to express this improvement per additional parameter and then divide by $\widehat{\phi}$ to account for overdispersion.

Equivalent calculation starting from binomial fits

Because the coefficient estimates and deviances are the same for the corresponding binomial and quasi-binomial fits, we can also start with the ordinary binomial models and explicitly supply the estimated dispersion when carrying out the F comparison.

```
bmod2 <- glm(cbind(survive,total-survive) ~ period,
             family=binomial, troutegg)
anova(bmod2, bmod, dispersion=phihat, test="F")
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ period
Model 2: cbind(survive, total - survive) ~ location + period
  Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
1        16     913.56
2        12      64.50  4   849.06 39.822 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Main messages from the example

1. An ordinary binomial GLM assumes

$$\mathrm{Var}(Y_i) = \frac{\mu_i(m_i - \mu_i)}{m_i},$$

corresponding to $\phi = 1$.

2. If the observed variability is larger than this, we may have **overdispersion**.
3. Quasi-binomial regression allows

$$\mathrm{Var}(Y_i) = \phi \frac{\mu_i(m_i - \mu_i)}{m_i}.$$

4. The mean model and coefficient estimates are unchanged, because the common factor ϕ cancels from the estimating equations for $\boldsymbol{\beta}$.
5. The inference changes because $\widehat{\phi}$ inflates the estimated covariance matrix and standard errors.
6. When comparing nested quasi models, we account for $\widehat{\phi}$ and use an F test rather than the usual χ^2 deviance comparison.